

MS DEV SUMMIT

AI for the Better (Life of people with Disabilities)

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www.msdevsummit.com



10th of December 2021



Online

ORGANIZER:

AcademicPartners
FUNDACJA



Hi, My name is!

- Dennie Declercq
- @DennieDeclercq
- #DreamingIsBelieving
- President & Developer DDSOFT





 Microsoft
Cognitive Services

And I am NOT an AI
engineer

PRIMER INTO ACCESSIBILITY



Kind of disabilities



Learning
disability



Autism



Visual



Auditory



Motoric



Much more...

Learning disability?

Significant limitations in both:

- Intellectual functioning
- Adaptive behavior

Learning Disability

Intellectual Functioning

Also called intelligence

Mental capacity

IQ score between 70-75

Adaptive Behavior

Conceptual Skills

- Language/literacy
- Money
- Time/numbers

Social Skills

- Social responsibility
- Self-esteem

Practical skills

- Activities daily living
- Personal care / healthcare



Principles of Inclusion

- Same activities
- Same district
- Work Together
- Adjustments par person

Digital Difficulties for PWD.

Learning Disability

- Read
- Understand

Autism

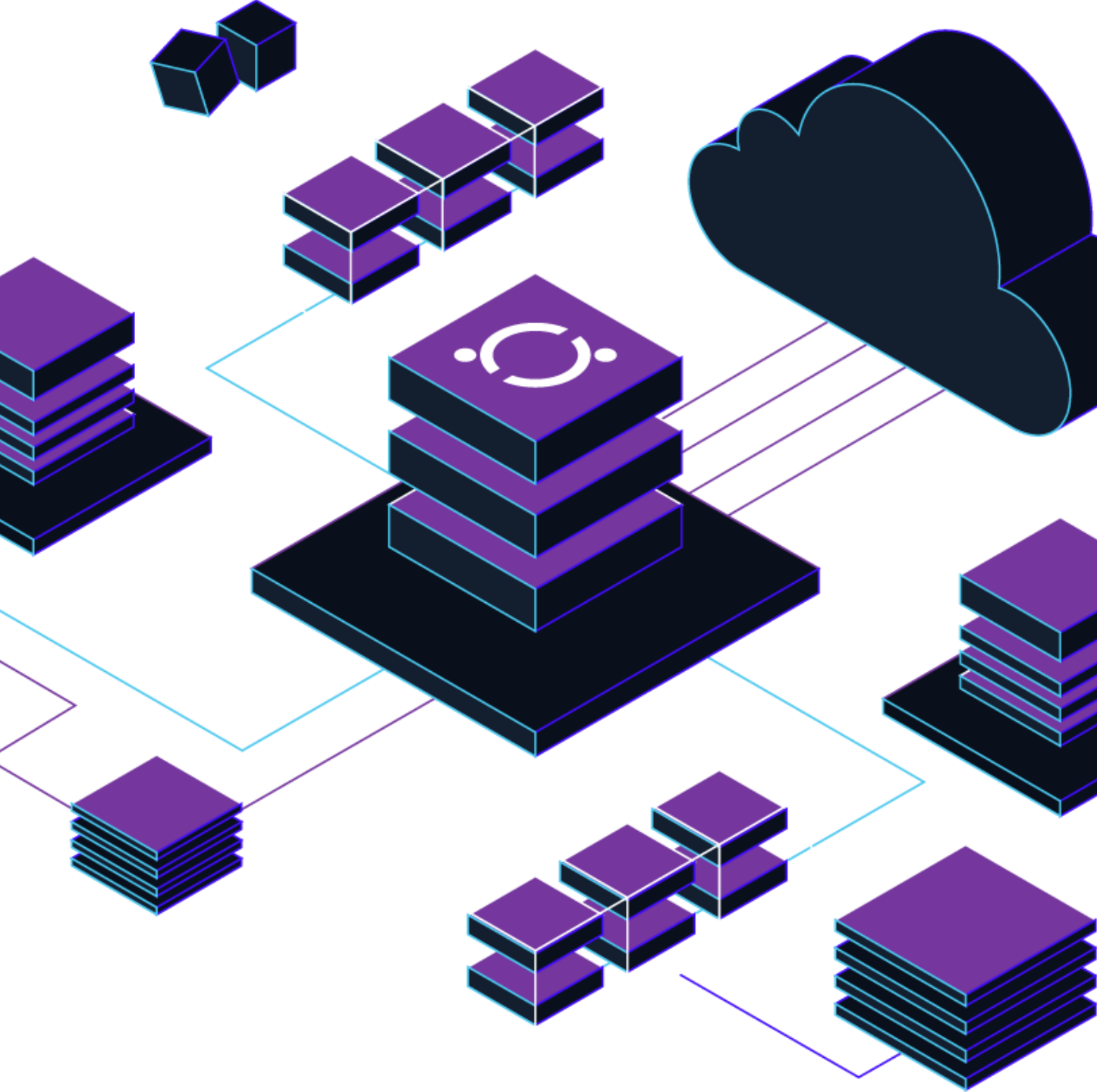
- Understand
- See structure

Blindness or vision loss

- Decreased ability to read
- Decreased ability to navigate

Auditory disabilities

- Hearing notifications
- Following movies without audio description.



Our Focus

- People with Learning disabilities
- Autism
- People with mental health situations

MICROSOFT COGNITIVE SERVICES API'S

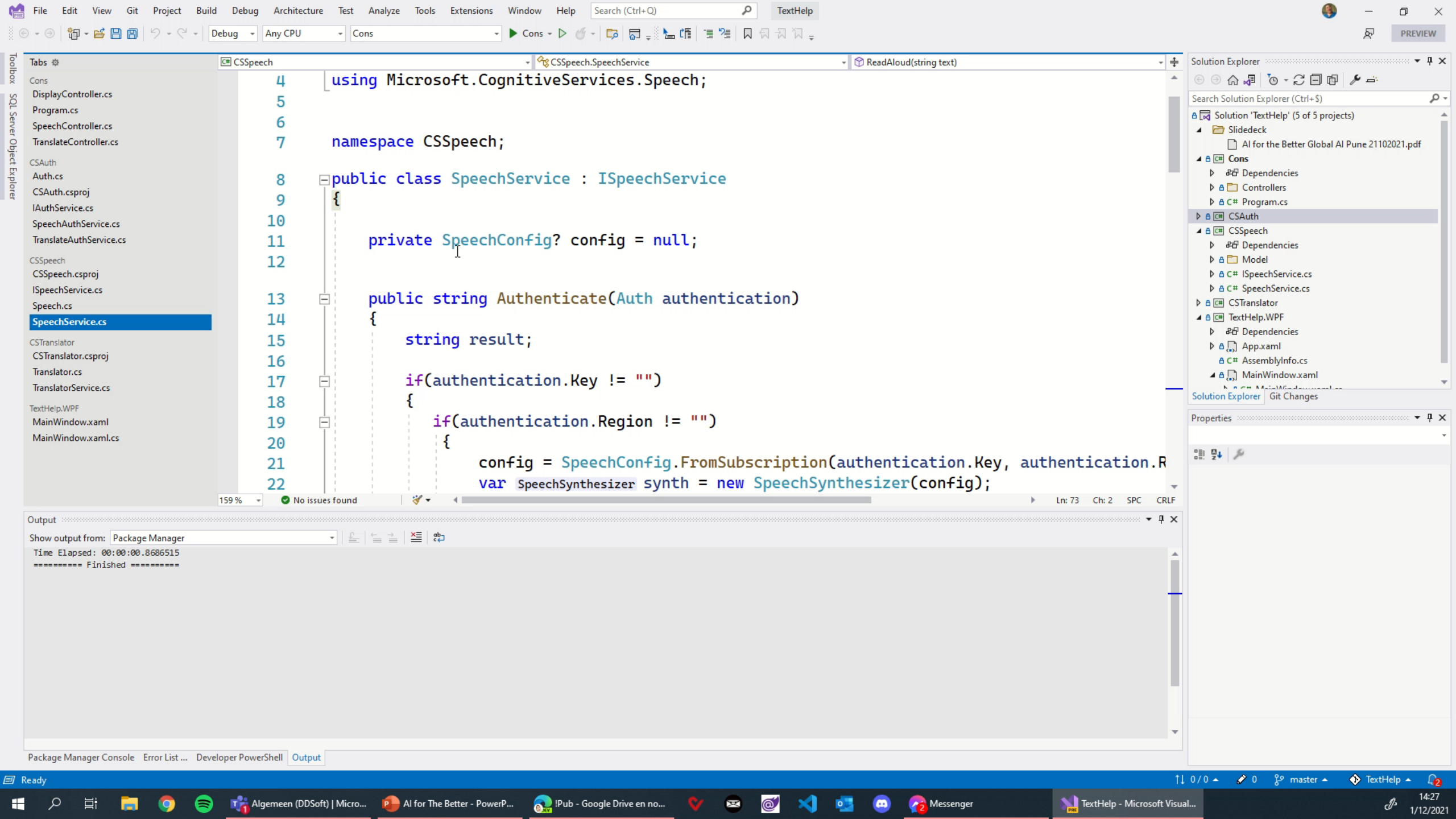
GETTING STARTED WITH COGNITIVE SERVICES

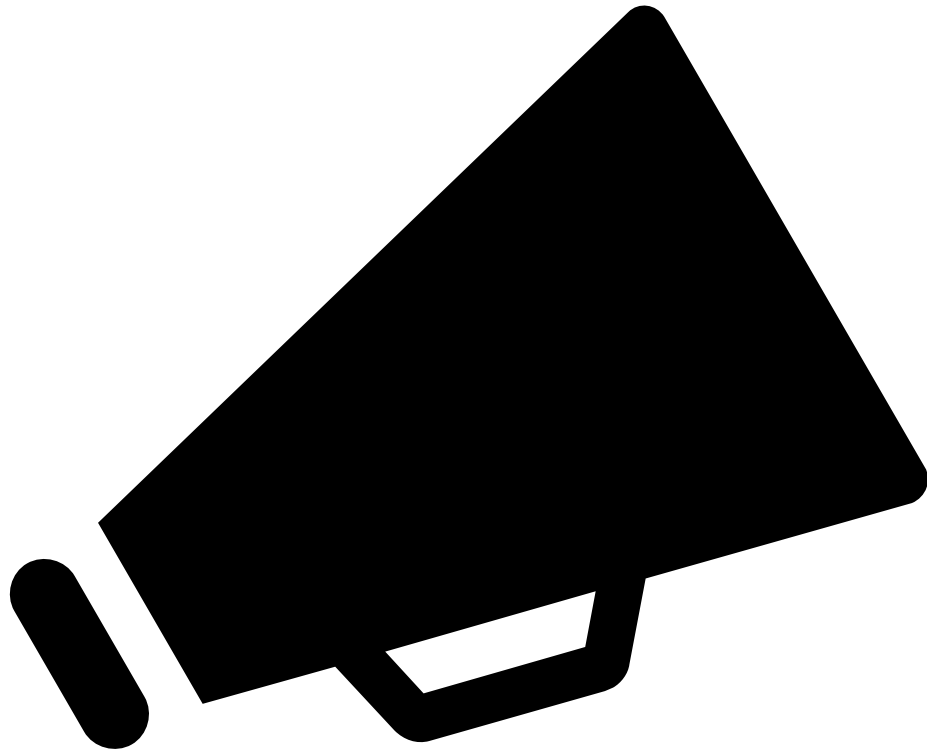
AI + Machine Learning > Cognitive Services

 Bing Autosuggest Microsoft Help users complete queries faster by adding intelligent type-ahead capabilities 	 Bing Custom Search Microsoft Create a custom search engine with ad-free results 	 Bing Entity Search Microsoft Recognize and classify named entities, and find search results based on them 	 Bing Search Microsoft Enable safe, ad-free search for your users with Bing, a leading search engine 	 Bing Spell Check Microsoft Help users identify and fix word breaks, slang, names, homonyms, and brands 	 Computer Vision Microsoft An AI service that analyzes content in images 	 Content Moderator Microsoft An AI service that detects unwanted content 
 Custom Vision Microsoft An AI service and end-to-end platform for applying computer vision to your specific scenario 	 Face Microsoft An AI service that analyzes faces in images 	 Speaker Recognition Microsoft An AI service that enables you to identify individual speakers or use speech as a means of verification 	 Speech Microsoft An AI service that enables you to embed human-like speech capabilities into your apps 	 QnA Maker Microsoft An AI service that enables you to create a bot from your data in minutes 	 Text Analytics Microsoft An AI service that enables you to unlock insights from natural language text using sentiment 	 Translator Text Microsoft An AI service that enables you to easily translate text between more than 60 languages 
 Personalizer Microsoft An AI service that delivers a personalized user experience 	 Language Understanding Microsoft Teach your apps to understand commands from your users 	 Ink Recognizer (Preview) Microsoft An AI service that recognizes digital ink content, such as handwriting, shapes, and document layout 	 Immersive Reader (preview) Microsoft An AI service that helps users read and comprehend text 	 Form Recognizer (preview) Microsoft The AI service that extracts information from documents 	 Anomaly Detector (preview) Microsoft An AI service that helps you foresee problems before they occur 	 Cognitive Services Microsoft Connect powerful AI to your apps 



DEMO: Creating a Cognitive Service on Azure



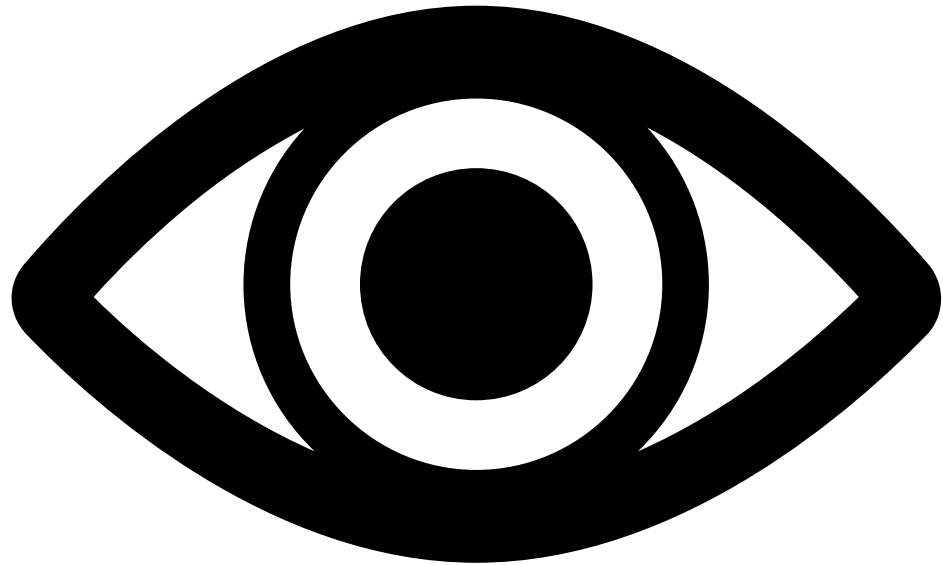


Speech Service

- Text-to-speech
- Custom Voice

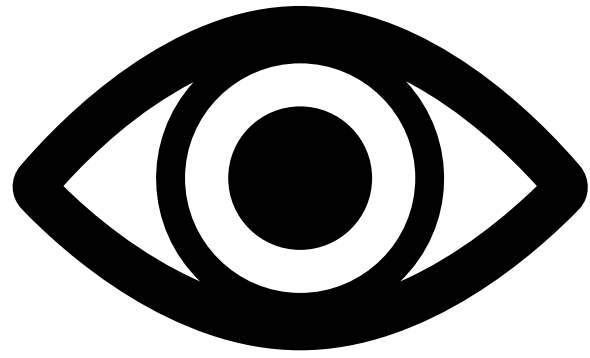


DEMO: Speech API



Vision API

- Analyzes images
- Let PWD take pictures and tells what it is.



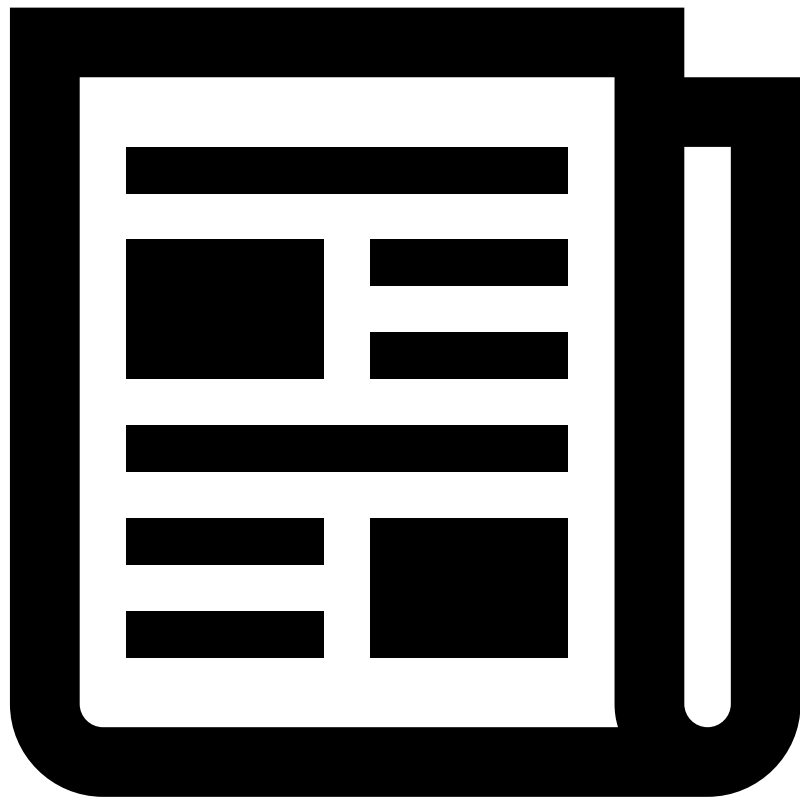
Face API

- Detect Emotion
- Detect Facial Hair
- Detect Smile
- Detect Age



BOT Service

- Provide BOTs
- BOTs really can assist PWD
- Make clear guidelines



Immersive Reader

- Accessibility features like:
 - Reading aloud websites
 - Linking pictographs to websites
 - Text appearance options



Translator Service

- Translation
- Language Detection



DEMO: Translator API

PROBLEMS

A thin, vertical white line is positioned to the right of the word "PROBLEMS", extending from the top of the word down to the bottom of the word.

Literacy

- Reading information
- Understanding written text
- Recognizing money
- Math



Executive Functions

- Planning
‘simple’ tasks
- Making choices
- Concentration



‘FEELING’

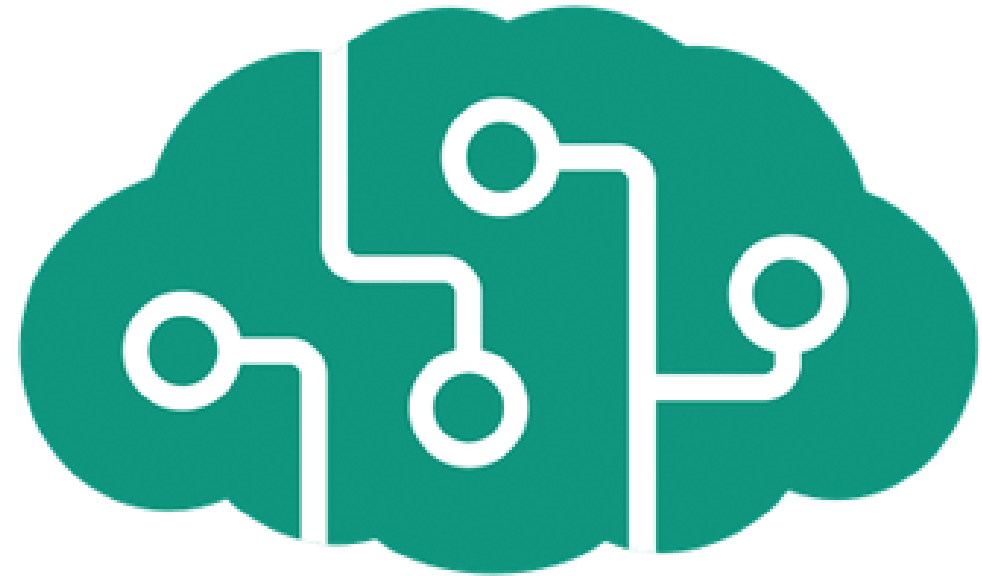
“I have no elbows, I don’t feel if I am going over your borders, and if this annoys you” -Dennie Declercq

Self-trust

- Self-trust can be hard
- Can I really do this?
- Will this really work?



AS YOU
SEE...



 Microsoft
Cognitive Services

THE REAL INSPIRATION





The Immersive Recipe



My vision barber

Custom Vision

Face API

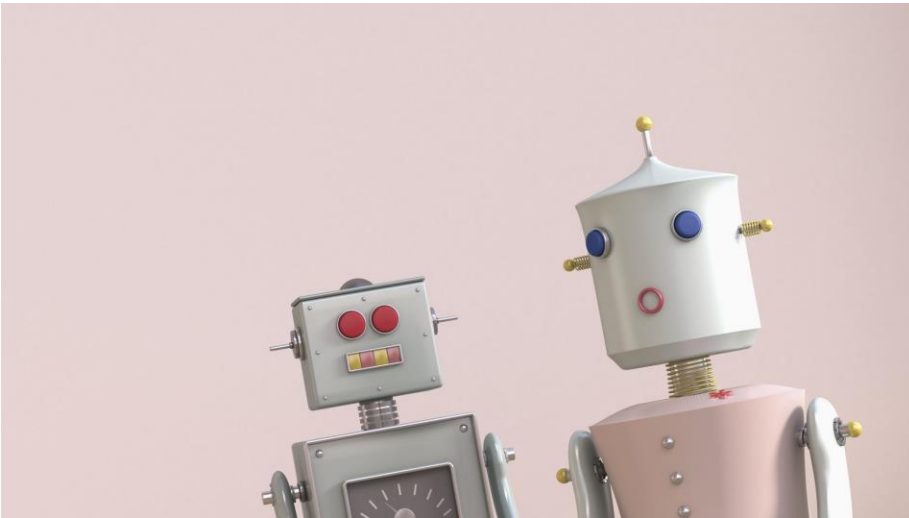
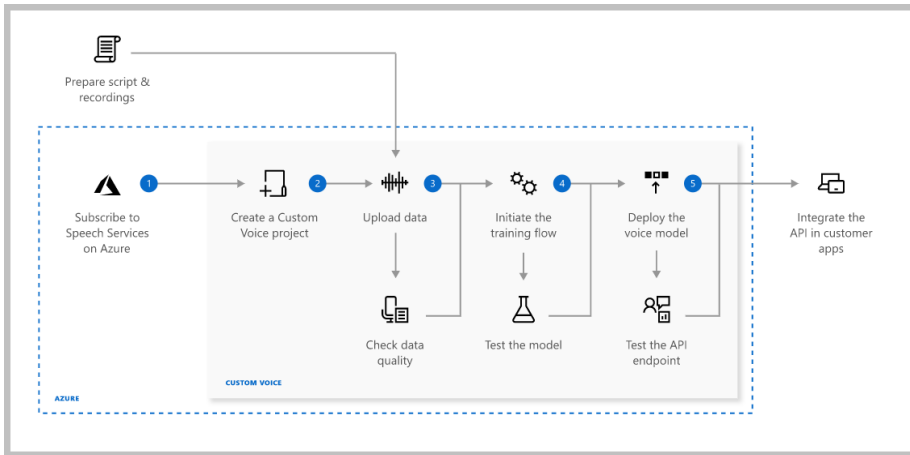
```
[Newtonsoft.Json.JsonProperty(PropertyName="beard")]  
public double Beard { get; set; }
```

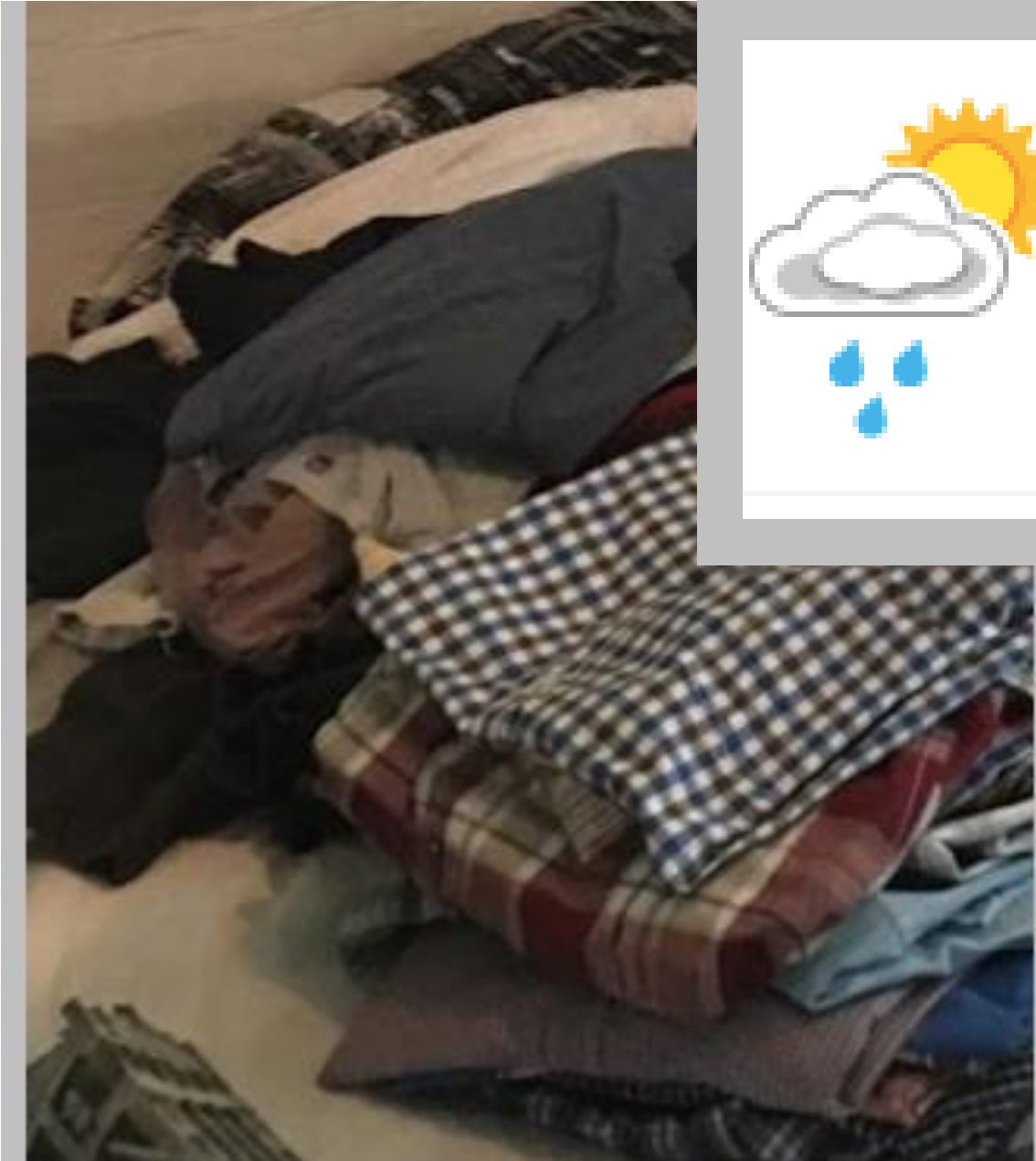
My Artificial Mom

Text To Speech

BOT Service

Custom voice





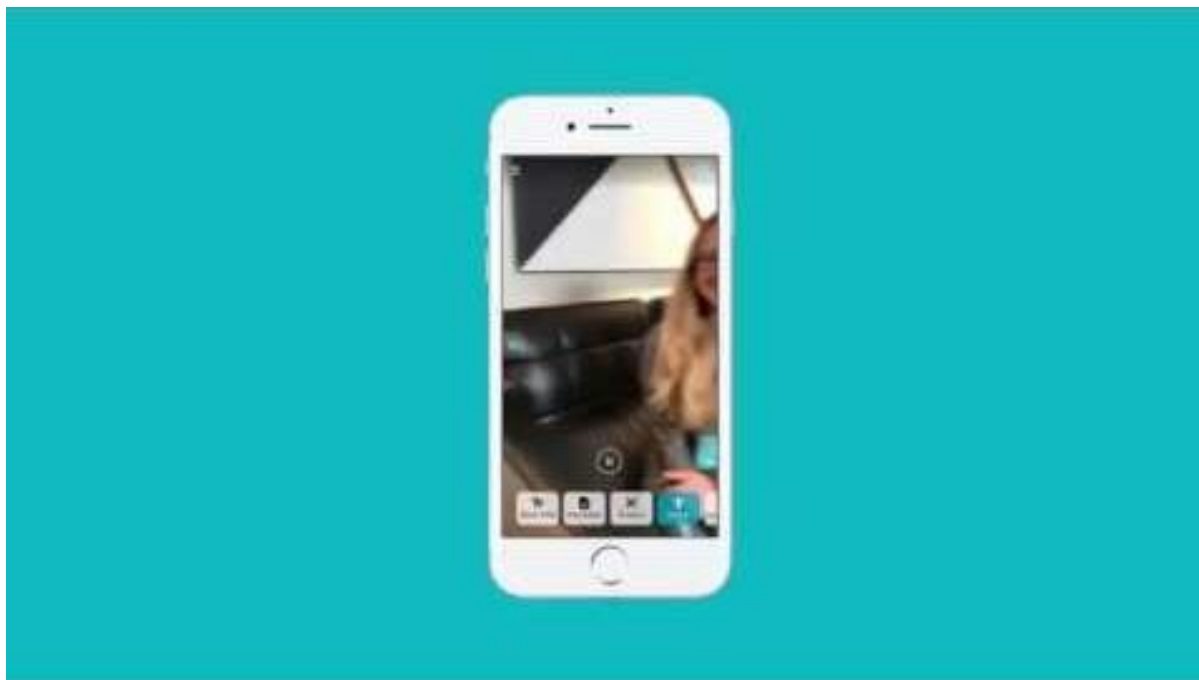
The AI wardrobe

Graph API

Computer Vision

REAL WORLD INSPIRATION

A thin, vertical white line is positioned to the right of the text, extending from the top of the word 'INSPIRATION' to the top of the word 'REAL'.



MICROSOFT SEEING AI
<https://go.ddsoft.be/msftseeingai>



MICROSOFT TRANSLATOR
<https://go.ddsoft.be/msfttranslator>



HELPICTO
<https://go.ddsoft.be/helpicto>

People can add
images or adjust
content as needed



THANK YOU POWERPOINT!
[HTTPS://GO.DDSOFT.BE/AIPPT](https://go.ddssoft.be/aippt)

A man with short brown hair and a beard, wearing a black t-shirt, is sitting at a long wooden table in a modern office. He is looking down at a laptop in front of him. On the table, there is also a black backpack, a clear plastic cup with a straw, and a black can. The background shows large windows with a view of a city and a wall with a colorful polka-dot pattern.

AI for Accessibility grants

Empower your work through grants, investments of technology, and expertise.

Apply for a grant >

<https://go.ddsoft.be/msftai>

CALL: AI FOR ACCESSIBILITY

Research	In accessibility / People with disabilities
Learn	Microsoft Cognitive Services
Get	Further inspired
Play	With existing tools

Getting
started

WRAP UP

Links & References

- Microsoft Seeing AI: <https://go.ddsoft.be/msftseeingai>
- Helpicto: <https://go.ddsoft.be/helpicto>
- Microsoft Translator movie: <https://go.ddsoft.be/msfttranslator>
- AI in PowerPoint: <https://go.ddsoft.be/aippt>
- Microsoft AI for Good: <https://go.ddsoft.be/msftai>
- Demo: 'TextHelp' on GitHub: <https://go.ddsoft.be/texthelp>



Thanks! | | Q&A

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- @DennieDeclercq
- #DreamingIsBelieving



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Thank you for watching
Remember to leave your **questions**
and **rate** the presentation
in the screen below.



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10th of December 2021



Online

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